

RaceDay — Part 1 API Endpoint Plan

This plan defines the REST API to be implemented in Part 2 and is aligned with the Part 1 database/ERD design.

Roles: Organiser manages events, categories, enrolments and results. Participant registers, browses events, enrolls by category and views personal history.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	`/api/auth/register`	Creates a new participant account.	Public	`{ fullName, email, password }`	`201 Created` with user ID/profile; `400` validation error; `409` email exists
POST	`/api/auth/login`	Authenticates a user and returns an access token.	Public	`{ email, password }`	`200 OK` with token and role; `401` invalid credentials
GET	`/api/users/me`	Returns the profile of the authenticated user.	Any logged-in user	None	`200 OK`; `401` unauthenticated
PUT	`/api/users/me`	Updates the authenticated user's profile.	Any logged-in user	`{ fullName, email }`	`200 OK`; `400` validation; `409` email exists
GET	`/api/events`	Lists upcoming RaceDay events.	Public	None	`200 OK` with event list
GET	`/api/events/{id}`	Returns details for one event, including its categories and route.	Public	None	`200 OK`; `404` event not found
POST	`/api/events`	Creates a new event.	Organiser	`{ eventName, description, eventDate, location, maxParticipants, categoryIds }`	`201 Created`; `400` invalid data; `403` forbidden
PUT	`/api/events/{id}`	Updates an existing event.	Organiser	Event fields to update	`200 OK`; `404`; `403`; `400`
DELETE	`/api/events/{id}`	Deletes an event that is no longer needed.	Organiser	None	`204 No Content`; `404`; `403`; `409` if related records prevent deletion
GET	`/api/categories`	Lists all event categories.	Public	None	`200 OK` with categories
POST	`/api/categories`	Creates a new event category.	Organiser	`{ categoryName, description }`	`201 Created`; `400`; `409` duplicate
PUT	`/api/categories/{id}`	Updates a category.	Organiser	`{ categoryName, description }`	`200 OK`; `404`; `403`
DELETE	`/api/categories/{id}`	Removes a category not in active use.	Organiser	None	`204 No Content`; `404`; `409` in-use category
GET	`/api/events/{eventId}/categories`	Returns categories available for a specific event.	Public	None	`200 OK`; `404` event not found
POST	`/api/events/{eventId}/enrolments`	Enrols the logged-in participant in an event using a selected category.	Participant	`{ categoryId }`	`201 Created`; `400` invalid category; `404`; `409` already enrolled/full
GET	`/api/enrolments/me`	Lists the logged-in participant's own enrolments.	Participant	None	`200 OK` with enrolments
GET	`/api/events/{eventId}/enrolments`	Views all enrolments for an event.	Organiser	None	`200 OK`; `403`; `404`
DELETE	`/api/enrolments/{id}`	Cancels the authenticated participant's enrolment.	Participant	None	`204 No Content`; `404`; `403`
GET	`/api/results/me`	Returns the authenticated participant's personal result history.	Participant	None	`200 OK` with results
GET	`/api/events/{eventId}/results`	Lists all results for an event.	Organiser	None	`200 OK`; `403`; `404`
POST	`/api/events/{eventId}/results`	Captures a participant result for an event.	Organiser	`{ enrolmentId, position, finishTime, distanceKm }`	`201 Created`; `400`; `404`; `409` if result exists
PUT	`/api/results/{id}`	Corrects an existing participant result.	Organiser	`{ position, finishTime, distanceKm }`	`200 OK`; `404`; `403`
DELETE	`/api/results/{id}`	Removes an incorrect result.	Organiser	None	`204 No Content`; `404`; `403`
GET	`/api/events/{eventId}/route`	Returns route information used to prepare for race day.	Public	None	`200 OK`; `404`
GET	`/api/events/{eventId}/weather`	Returns latest weather information for an event.	Public	None	`200 OK`; `404`

Access-Control Rules

1. Authentication endpoints are public. 2. Event/category browsing is public. 3. Organiser-only operations reject Participants with 403. 4. Participant-only operations enforce the Participant role. 5. /me endpoints only expose the authenticated user's own data. 6. JSON is used for requests/responses. 7. Use 400 for validation, 401 for unauthenticated access, 403 for forbidden access, 404 for missing resources and 409 for conflicts.

Database Mapping

Users, Events, Categories, EventCategories, Routes, Enrolments, Results and Weather.